Experiment 5

Implementation of RSA Algorithm

Aim

To implement the RSA algorithm for secure encryption and decryption of plaintext using public and private keys.

Description

RSA (Rivest–Shamir–Adleman) is an asymmetric encryption algorithm that uses two keys: a public key for encryption and a private key for decryption. It is widely used for secure communication and digital signatures. The security of RSA is based on the difficulty of factoring large prime numbers.

Algorithm

1. Start the program.
2. Enter two prime numbers p and q.
3. Calculate $n = p \times q$.
4. Calculate $\varphi(n) = (p - 1) \times (q - 1)$.
5. Enter the public key e.
6. Find the private key d such that $(d \times e) \bmod \varphi(n) = 1$.
7. Enter the plaintext as a number less than n.
8. Encrypt using C = (M^e) mod n.
9. Decrypt using M = (C^d) mod n.
10. Display the encrypted and decrypted values.
11. Stop the program.

Code

```
p = int(input("Enter prime number p: "))

q = int(input("Enter prime number q: "))

e = int(input("Enter public key e: "))

n = p * q

phi = (p - 1) * (q - 1)
```

```
for i in range(2, phi):
    if (e * i) % phi == 1:
        d = i
        break
m = int(input("Enter Plain Text (number): "))
c = (m ** e) % n
print("Encrypted Text:", c)
m = (c ** d) % n
print("Decrypted Text:", m)
```

Output: